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Research Project Report on the Topic:

Automatic Generation of Cover Songs Using Deep Learning

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Annotation

Automatic cover song generation solves the problem of producing a new stylistic version of an existing musical track while preserving its recognizable musical content. This makes the task substantially harder than unconditional music generation, especially in the audio domain, where content and style are closely connected. This work studies cover generation as a music style transfer problem and focuses on the challenge posed by multi-instrument tracks. The proposed approach is based on a diffusion model operating in latent space, where musical structure and style are disentangled and then recombined in a conditional denoising U-Net. To improve style transfer for complex mixtures, the input audio is first separated into instrumental stems, and each of them is then processed individually before conditioning the diffusion model. We compare the original pipeline, a stem-only extension, and a mixed variant that combines separated stems with the original mixture. The quantitative results do not show a clear improvement over the baseline: the stem-only setting performs worst, while the mixed setting remains close to the original model. At the same time, listening suggests that the proposed extension can produce a softer and in some cases more appropriate timbral transformation, although the generated audio often becomes more muffled.

Аннотация

Автоматическая генерация каверов решает задачу создания новой стилистической версии уже существующего музыкального трека при сохранении его узнаваемого музыкального содержания. Эта задача заметно сложнее, чем задача безусловной генерации музыки, особенно в аудиодомене, где содержание и стиль тесно переплетены. В данной работе генерация каверов рассматривается как задача переноса музыкального стиля, причем особое внимание уделяется трудностям, возникающим при работе с мультиинструментальными треками. Предлагаемый подход основан на диффузионной модели, работающей в латентном пространстве, где структура и стиль аудио раскладываются на независимые представления, а затем объединяются в условной denoising U-Net. Для более точного переноса стиля в сложных миксах исходное аудио сначала разделяется на инструментальные дорожки, после чего каждая из них обрабатывается отдельно перед подачей условий в диффузионную модель. В работе сравниваются три варианта: исходный пайплайн, вариант только с инструментальными дорожками и смешанный вариант, сочетающий отдельные дорожки с исходным миксом. По количественным метрикам заметного улучшения по сравнению с базовым методом получить не удалось: вариант только с инструментальными дорожками оказался худшим, а смешанный режим показал результаты, близкие к исходному пайплайну. При этом на

слух предложенное расширение в отдельных случаях дает более мягкий и уместный тембр, хотя в целом сгенерированный звук нередко становится более глухим.

Keywords

Conditional Music Generation, Cover Song Generation, Music Style Transfer, Deep Learning, Diffusion Models, Source Separation

1 Introduction

The history of computer music began in the 1950s with the first attempts at automating composition using large mainframe computers [1]. Since then, the field has evolved from simple Markov chains [2] to modern neural architectures [3, 4, 5, 6] capable of synthesizing realistic audio directly from raw waveforms or compressed latent representations. In the modern world, generative models are increasingly used as tools for supporting creativity: they help authors overcome creative blocks, prototype new musical ideas, and automate routine parts of music production [8]. One natural application of these technologies is automatic cover generation. In musicology, a cover is usually understood as a re-recording or reinterpretation of a previously released song [7], often involving noticeable changes in style or arrangement. Traditionally, creating a convincing cover still requires knowledge of music theory, instrumental performance skills, and studio processing experience. Automating this process could make stylistic reinterpretation of music much more accessible.

The main challenge is to build a system that can transform an existing musical recording into a cover version in another genre or style. The output audio should remain recognizable as a reinterpretation of the original: it should preserve melody, harmonic progression, rhythmic feel, and overall song structure, while also adapting instrumentation, timbre, and texture to the target style. A natural way to formulate this problem is to separate music into two components: content, which includes the invariant musical foundation, and style, which includes the sonic and performative characteristics of the recording. Under this view, cover generation becomes a neural style transfer problem in which the model re-renders the musical content of the source recording with the stylistic properties of a target reference. The system takes as input the original audio and a target style reference and must produce a new recording that jointly maximizes content preservation and style authenticity.

Although this formulation is natural, adapting neural style transfer concepts from the image domain [9] to audio introduces substantial difficulties. Early audio-domain approaches attempted to apply image-style-transfer techniques directly to spectrograms [12, 13], effectively treating time and frequency as if they were interchangeable spatial axes. This turned out to be a serious limitation: in audio, these axes play different perceptual roles, and handling them symmetrically often damages musical structure and introduces audible artifacts [14, 15]. As a result, direct spectrogram-based style transfer has only limited success for realistic cover generation.

Recent progress in large neural generative models, especially transformers and diffusion models, has led to impressive results in text-conditioned music generation [3, 4, 5, 6]. Nevertheless, concepts such as timbre, musical style, and genre boundaries are often difficult to specify precisely with text alone. This motivates an alternative paradigm in which the desired style is provided not by text, but by another audio

example [16, 17, 18].

In this work, we follow this direction and investigate whether source separation can improve audio-reference cover generation within a latent diffusion pipeline. More specifically, we study whether separating a mixture into stems before using them for conditioning can make style transfer more accurate for multi-instrument music.

2 Related Work

2.1 Symbolic Methods

Early research on automated music arrangement and generation predominantly operated in the symbolic domain, using representations such as note sequences, chord symbols and MIDI (Musical Instrument Digital Interface) tracks [10]. Systems, such as Flow Machines [11], modeled harmonic and melodic patterns with statistical methods and were useful for capturing high-level musical regularities. However, these approaches are inherently disconnected from the continuous domain of audio. They require either symbolic input, which means that real recordings must first be transcribed, or an additional synthesis stage to return to waveform space. Both steps can introduce substantial errors and reduce realism. For this reason, symbolic methods are less suitable for the specific task studied here, namely transforming one audio recording into another while preserving recognizable musical identity.

2.2 Audio-domain methods

The success of image neural style transfer [9] motivated its direct application to audio spectrograms. Initial works [12, 13] used Gram-matrix-based objectives to transfer texture-like characteristics between sounds. However, such visual features proved suboptimal for audio and often produced noisy outputs with audible artifacts. This led to audio-specific adaptations, including shallow convolutional networks [19], temporal convolutions [15], and rhythm-aware processing schemes [20], all designed to preserve spectro-temporal structure more carefully. Even so, optimization-based spectrogram transfer remained limited by weak long-range structure control.

To move beyond direct spectrogram optimization, later work turned to learned latent representations. Autoencoder-based approaches [21, 22] showed that compact latent spaces can separate content and style more explicitly and can therefore provide a better basis for controllable generation. This observation is especially important for cover generation, because it suggests that style transfer may be easier in a learned latent representation than in raw waveform or spectrogram space.

In parallel, large generative models have substantially improved music synthesis quality. MusicLM [3], MusicGen [4], and latent diffusion systems such as AudioLDM [5] and Moûsai [6] demonstrate that modern generative models can produce increasingly realistic audio. However, most of these systems are primarily designed for text-conditioned generation rather than for transferring the sonic characteristics of one recording onto the musical structure of another.

The most relevant recent works for our setting are diffusion-based methods for music style transfer. Li et al. [16] propose an inversion-based approach with time-varying prompts extracted from an audio reference, but the method still struggles with complex multi-instrument content. Kim et al. [17] introduce a training-free latent diffusion transfer method, yet the generated output often fails to preserve the original melodic and harmonic structure. Demerlé et al. [18] present a framework with separate timbre and structure encoders, trained with an adversarial criterion to encourage disentanglement. This method is the closest to our setting, but it still operates on a single mixture-level representation. Our work can therefore be seen as an extension of this pipeline, where source separation is introduced in an attempt to provide more precise conditioning for different instrumental components.

3 Method

3.1 Base pipeline

Large supervised datasets of the form

$$\left(x^{\text{content}}, x^{\text{style}}, x^{\text{cover}}\right)$$

are essentially unavailable for cover generation. Because of that, the method of Demerlé et al. [18] does not learn a direct mapping from input tracks to target covers. Instead, it learns to decompose an audio example into two representations: a time-varying structure code and a global style code, and then uses these codes to condition a latent diffusion decoder.

Let x denote an input waveform and let Enc and Dec be the encoder and decoder of a pretrained audio autoencoder. The waveform is first mapped to a latent representation

$$z_0 = \text{Enc}(x),$$

and the diffusion model is trained in this latent space rather than in waveform space. This reduces computational cost and makes denoising more stable [18, 23]. A temporal structure encoder E_ψ^S receives a CQT representation of the source audio and produces a time-dependent code v_S. A timbre encoder E_ϕ^T

receives the latent audio representation and produces a global semantic style code v_T.

Following the latent diffusion formulation, noisy latent variables are sampled as

$$z_t = \sqrt{\alpha_t}\, z_0 + \sqrt{1-\alpha_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I),$$

where $\alpha_t = \prod_{s=1}^{t}(1-\beta_s)$ is determined by the noise schedule [24, 25]. The denoising network is a 1D U-Net with residual blocks and self-attention, and it is trained to predict the clean latent from the noisy input and the conditioning codes:

$$L_{\text{diff}} = \mathbb{E}_{t,z_0,\varepsilon} \left\| D_\theta(z_t, v_S, v_T, t) - z_0 \right\|_2^2 . \tag{1}$$

This means that the diffusion model learns to resynthesize the target audio in latent space while the autoencoder decoder Dec converts the denoised latent back to waveform.

The central modeling assumption is that structure and style should be statistically independent enough to be controlled separately. To encourage this, Demerlé et al. [18] introduce an adversarial criterion. A discriminator D_ζ tries to predict the style code from the structure code:

$$L_D = \mathbb{E}_{v_S,v_T} \left\| v_T - D_\zeta(v_S) \right\|_2^2 .$$

The encoders and the diffusion decoder are then trained with the objective

$$L_{\text{total}} = L_{\text{diff}} - \gamma L_D,$$

while the discriminator itself minimizes L_D. Intuitively, the discriminator should fail to recover style from the structure representation; if this happens, the model is forced to encode style and content in different variables.

Conditioning is incorporated into the denoising U-Net exactly as in the original paper. Structure is passed through channel concatenation: at each block, projected structure features are appended to the current feature map. Style controls the hidden representation through learned rescaling and shifting. If h_l is the hidden representation at layer l, then

$$h_l^{\text{style}} = r_l(v_T) \odot h_l + s_l(v_T),$$

where $r_l(\cdot)$ and $s_l(\cdot)$ are outputs of a multilayer perceptron (MLP) applied to the semantic style code and $\odot$ denotes elementwise multiplication.

3.2 Source-separation extension

The mixture-level conditioning of the baseline method is a natural limitation for multi-instrument music. A single global timbre code has to simultaneously describe drums, bass, guitars, keyboards, and other instrumental components. This is especially problematic when the desired genre transformation affects different sources differently. For example, in a metal-to-jazz transfer, distorted guitars, drums, and bass should not be stylized in the same way.

Source separation could help for several reasons. First, in a full mixture the style encoder has to compress many different sources into a single vector, so stronger instruments may hide weaker but still musically important components. Second, disentangling content and style is easier at the stem level: within a narrower source class, the encoder can focus on a cleaner signal and the adversarial criterion has a simpler task than in the case of a dense mixture. Third, many genre transformations are inherently source-specific. Drums may need a different rhythmic texture, bass may need a different articulation, and harmonic instruments may need a different timbral envelope. A stem-wise representation could therefore allow the model to attach stylistic changes to the appropriate instrumental component instead of averaging them over the whole mix. Finally, the mixed setting keeps the original mixture as an additional conditioning input, which may preserve global coherence while still letting the stem branches provide more detailed source-level information.

To address this issue, we introduce source separation before the disentanglement stage. Let x be a multi-instrument mixture and let Demucs [26] decompose it into K stems

$$x \approx \sum_{k=1}^{K} s_k.$$

In the stem-only setting, the conditioning set consists of the separated stems $\{s_1, \ldots, s_K\}$. In the mixed setting, we additionally keep the original mixture as one more stream $s_0 = x$, so the conditioning set becomes $\{s_0, s_1, \ldots, s_K\}$.

For each stream s_k, we compute a structure representation from the content audio and a style representation from the style audio:

$$v_{S,k} = E^{S}_{\psi_k}\big(\mathrm{CQT}(s_k^{\text{content}})\big), \qquad v_{T,k} = E^{T}_{\phi_k}\big(\mathrm{Enc}(s_k^{\text{style}})\big).$$

Each stream has its own trainable copies of the structure and timbre encoders, but all these copies are initialized with the pretrained single-stream weights of Demerlé et al. [18]. This makes the extension a fine-tuning problem rather than training a large model with separate stem branches from scratch.

The stem-conditioned denoising objective keeps the same structure as equation (1), but now the

U-Net receives sets of structure and timbre codes:

$$L_{\text{diff}}^{\text{SS}} = \mathbb{E}_{t,z_0,\varepsilon} \left\| D_\theta\big(z_t, \{v_{S,k}\}_{k=0}^{K}, \{v_{T,k}\}_{k=0}^{K}, t\big) - z_0 \right\|_2^2 .$$

The adversarial term is applied independently to each stream,

$$L_D^{\text{SS}} = \sum_{k=0}^{K} \mathbb{E}_{v_{S,k}, v_{T,k}} \left\| v_{T,k} - D_{\zeta_k}(v_{S,k}) \right\|_2^2 ,$$

and the total training objective becomes

$$L_{\text{total}}^{\text{SS}} = L_{\text{diff}}^{\text{SS}} - \gamma L_D^{\text{SS}} .$$

Inside the denoising U-Net, content conditioning is still passed by concatenation:

$$\widetilde{h}_l = \text{Concat}\big(h_l, v_{S,0}, \ldots, v_{S,K}\big).$$

Style conditioning still uses learned rescaling and shifting, but now separately for each stream:

$$h_{l,k}^{\text{style}} = m_{l,k}(v_{T,k}) \odot \widetilde{h}_l + a_{l,k}(v_{T,k}).$$

In our implementation, the stream-wise modulated activations are averaged inside the block,

$$h_l^{\text{style}} = \frac{1}{K+1} \sum_{k=0}^{K} h_{l,k}^{\text{style}} ,$$

which preserves the conditioning mechanism of the original paper while allowing different streams to contribute different style transformations.

Figure 1 summarizes the full pipeline. The key hypothesis is that separating the mixture into stems makes the content/style decomposition easier and more precise, because each encoder copy can specialize to a narrower instrumental subproblem.

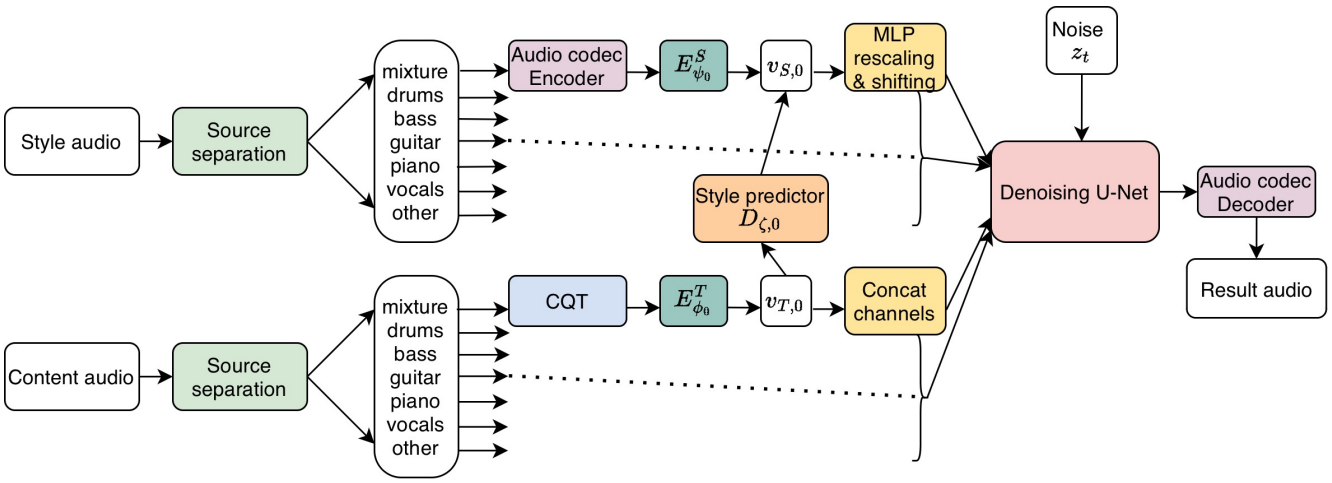

Figure 1: Proposed cover generation pipeline. Both the original mixture and the separated stems can be used for conditioning. Structure features are passed to the denoising U-Net by channel concatenation, while style features rescale and shift hidden activations using coefficients predicted by a multilayer perceptron (MLP) from semantic style codes.

4 Experiments

4.1 Dataset and preprocessing

We used the SLAKH dataset [27], a large collection of multitrack instrumental recordings suitable for controllable music generation and source-separation research. We use SLAKH because it provides multi-instrument arrangements with multiple instrument families, which makes it a relevant benchmark for testing whether stem-wise conditioning helps on realistic mixtures.

The dataset was preprocessed in two stages. First, for the experiments with source separation, each mixture was decomposed into stems with a pretrained Demucs model [26]. Second, because the diffusion model operates in the latent space of a pretrained audio autoencoder, we precomputed the encoded latent representations for all training excerpts. This reduced training cost. In the mixed setting, the original mixture was kept as an additional conditioning input alongside the separated stems.

4.2 Training protocol and compared settings

We consider three experimental settings:

1 **Without source separation.** This is the baseline pipeline of Demerlé et al. [18], where the whole mixture is processed as a single input and the model learns one structure code and one style code per example.

2 **Source separation only.** The mixture is decomposed into stems and only the stem-wise streams are used for conditioning.

3 **Source separation + original mixture.** The model receives both the separated stems and the original mixture as an additional stream.

For settings (2) and (3), we copied the pretrained weights released by the authors of the baseline method into all stem-specific encoder branches and then fine-tuned the resulting model with separate stem branches. This was important because training the full architecture from scratch would be computationally impractical in our setting.

Training was performed on an NVIDIA RTX 5090 GPU with 32 GB of memory. Depending on the experiment, the models were trained for roughly 50 000 to 200 000 update steps. This is substantially shorter than in the original paper, where the diffusion model was trained for 500 000 steps [18]. Therefore, our experiments should be interpreted as a fine-tuning study under limited compute rather than a full-scale reproduction of the original training regime.

4.3 Evaluation metrics

To compare the three settings, we used two simple proxy metrics designed for fast internal comparison of content preservation and style transfer quality.

Let x_c be the content audio, x_s the style audio, and x_r the generated result. The content preservation metric is computed from Mel-frequency cepstral coefficient (MFCC) features. If $M_c, M_r \in \mathbb{R}^{d\times T}$ are the MFCC matrices of x_c and x_r after length alignment, then

$$d_{\text{content}} = \frac{1}{T^2}\sum_{i=1}^{T}\sum_{j=1}^{T}\left\|M_c^{(:,i)} - M_r^{(:,j)}\right\|_2 .$$

Lower values indicate better preservation of the content signal.

The style transfer metric is computed from magnitude short-time Fourier transform (STFT) spectrograms. Here $|\text{STFT}(\cdot)|$ denotes the elementwise magnitude of the complex STFT. If $S_s = |\text{STFT}(x_s)|$ and $S_r = |\text{STFT}(x_r)|$, and

$$G(S) = \frac{SS^\top}{|S|},$$

where $G(S)$ is the Gram matrix of the spectrogram and $|S|$ is the number of elements in the matrix S, then the style metric is

$$d_{\text{style}} = \text{mean}\big((G(S_s) - G(S_r))^2\big).$$

In simpler terms, we first convert the style reference and the generated audio into magnitude spectrograms, then summarize each spectrogram with a Gram matrix. This Gram matrix captures broad frequency relationships over time, which makes it a rough proxy for timbre and overall style. We then

measure how different the two Gram matrices are. If the generated audio is stylistically closer to the target reference, this difference should be smaller. Again, lower values are better. Although these metrics do not replace human listening tests, they are sufficient for comparing the relative behavior of the three settings used in this work.

5 Results

Table 1 summarizes the outcome of the three experimental settings. No substantial quantitative gain was observed. The stem-only model, that is, the variant with source separation but without the original mixture, is clearly the weakest: both metrics are higher, which means that it preserves content worse and matches the target style worse. In contrast, the mixed setting, which combines separated stems with the original mixture, yields values close to the baseline without source separation.

Table 1: Relative comparison of the three experimental settings. Both metrics are minimized, so lower values are better.

Setting	Content preservation ↓	Style transfer ↓
(1) Without source separation	196.129	$8.8 \cdot 10^{-4}$
(2) Source separation only	249.485	$9.6 \cdot 10^{-4}$
(3) Source separation + original mixture	196.830	$8.9 \cdot 10^{-4}$

The qualitative picture is consistent with the table. Setting (2) gives the weakest audio result: the output sounds less coherent, instrumental balance is poorer, and the transfer often feels unstable. In many cases, it mainly focuses on the rhythmic part while suppressing the melody. Settings (1) and (3) are much closer to each other in overall quality. At the same time, they do not produce identical audio. In particular, for content audio with a metal character and style audio with a jazz character, the original baseline tends to keep a more audible distorted-guitar trace, whereas our stem-aware implementation often produces a noticeably softer sound. In this particular example, the proposed pipeline sounds more appropriate, but this remains a case-by-case observation rather than a general quantitative improvement.

A recurring problem across experiments is that the generated audio becomes more muffled. There are several possible reasons for this effect. First, source separation itself may suppress high-frequency details, especially in difficult mixtures. Second, the pretrained weights were learned for mixture inputs, which as a whole tend to sound brighter than individual stems. Therefore, the weights used to initialize the stem branches may have been at an unsuitable scale for stem-wise inputs. Third, the model with separate stem branches is harder to optimize than the baseline, while our training budget is smaller than the one used in the original paper. As a result, the model may produce audio that keeps the general structure of the sound but loses brightness and attack.

6 Conclusion

This work examined whether source separation can improve audio-domain cover generation in the latent diffusion pipeline of Demerlé et al. [18]. The proposed extension decomposes mixtures into stems, allocates separate encoder copies to each stream, and conditions the denoising U-Net with stem-wise structure and style representations. The idea is natural for multi-instrument music, because different instruments may need different stylistic transformations.

In practice, however, this extension did not produce an overall improvement. The stem-only variant performed substantially worse than the baseline, while the mixed variant, which additionally preserves the original mixture as a global conditioning stream, remained close to the baseline but did not clearly surpass it. The most likely reasons are separation artifacts, the loss of cross-instrument interactions when stems are processed independently, the increased complexity of the conditioning mechanism, and the smaller amount of fine-tuning compared with the original training setup. Therefore, the main conclusion of this study is negative but informative: naive stem-wise conditioning is not sufficient by itself to improve the quality of diffusion-based cover generation on multi-instrument music.

7 Future Work

Several directions could make this line of research more promising. Ideally, one would hire musicians and create a dedicated annotated dataset of song-cover pairs recorded under controlled conditions. Such a dataset would make it possible to study cover generation in a genuinely supervised setting instead of relying only on disentanglement and indirect conditioning. Beyond that, the separator and the generative model could be trained jointly, or at least adapted to each other, so that separation artifacts are not treated as fixed input noise. The current way of combining stems is intentionally simple; replacing it with cross-attention, hierarchical conditioning, or explicit interactions between stems may preserve ensemble coherence better than independent branches. Longer training and larger-scale fine-tuning are likely necessary, since the model with separate stem branches has many more effective degrees of freedom than the baseline. Perceptual losses and structured human evaluation could provide a more faithful picture than the lightweight metrics used here. Finally, it would be useful to test the method on better and more diverse datasets.

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